



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SEMESTER I, SESSION 2024/2025

BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECV 2113 – HUMAN COMPUTER INTERACTION

EV CHARGING STATIONS TRACKER

TASK 4

GROUP: NOBODY

GROUP MEMBERS	MATRIC NO
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KOO XUAN	A23CS0300
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Lecturer's name : Dr. Nur Zuraifah Syazrah binti Othman

Introduction

In project 4, we conducted our prototyping and evaluation of the BUT U app. BUT U revolutionize the charging experience by allowing the users to effortlessly find the charging station nearby and make the charging process, with payments seamlessly handled through an mobile-app linked to cashless payment. The three main tasks that users need to perform are: 1. Find a nearby charging station 2. Make payment 3. Track charging progress.

We had created a high-fidelity prototype using Figma, incorporating the wireframes and design ideas from project 3. Each team member was assigned different features of the BUT U app to complete the prototypes on time to allow us to move on to the next process which is user testing and evaluation.

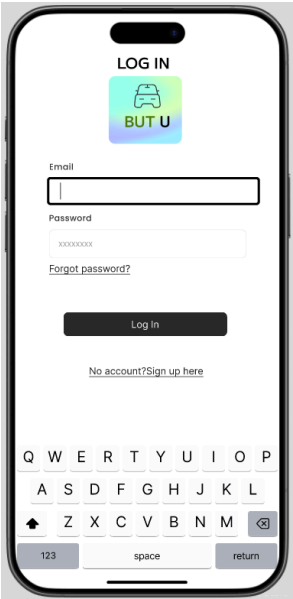
In the testing and evaluation process, we prepared briefing notes for our 3 target users who are the EV car owner, the company representative and the last is the traveler. Then, we asked the users to use the app in a charging station to charge their EV car and assisted them if they faced any difficulties. Lastly, we noted our observations, user feedbacks and the problems users faced to provide solutions.

Testing for User 1 (EV car owner) was handled by Tan Zhao Hong.

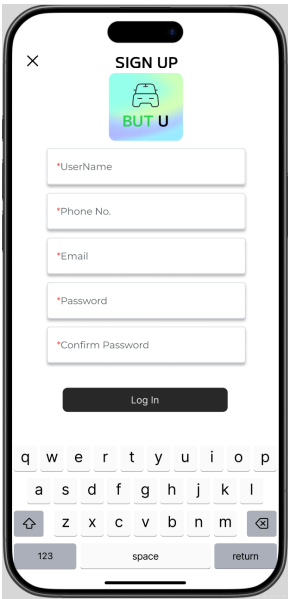
Testing for User 2 (Company) was handled by Nazatul.

Testing for User 3 (Traveler) was handled by Chin Pei Wen.

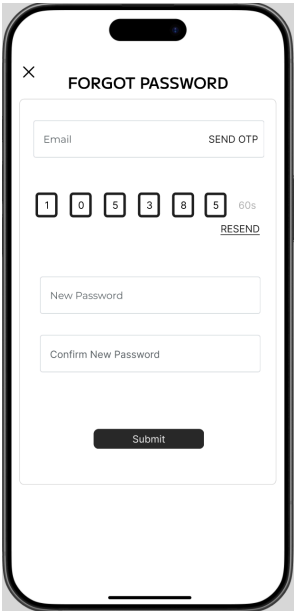
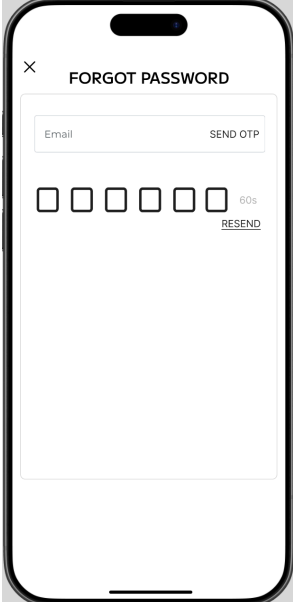
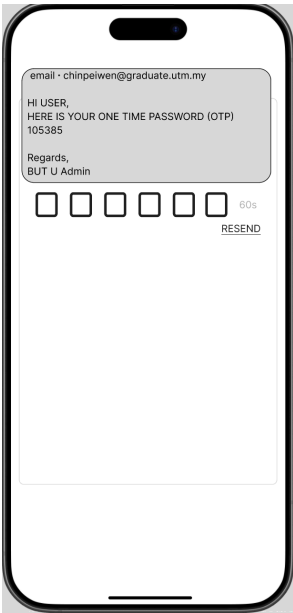
Screenshots of your prototype



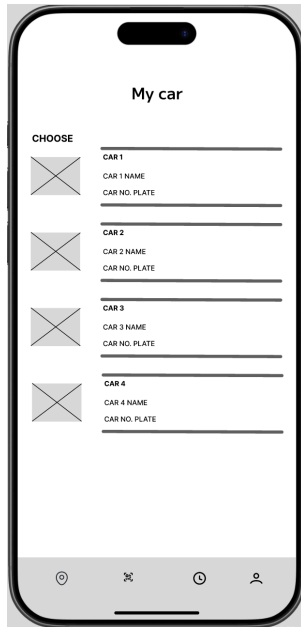
Login page



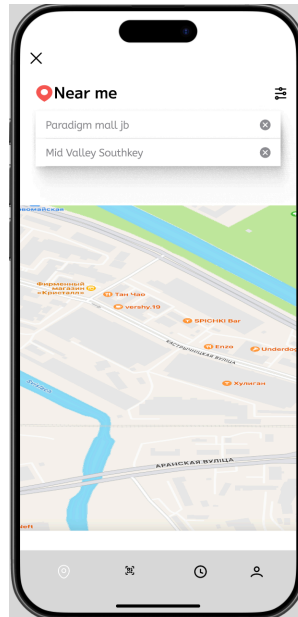
Sign up page



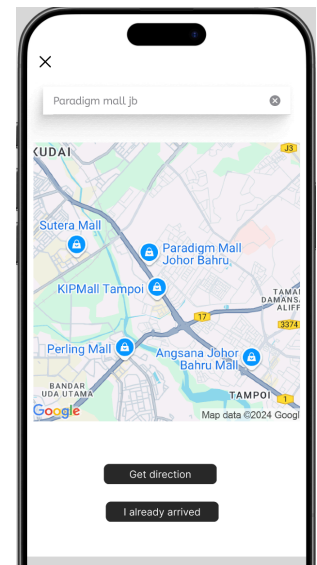
Forgot password pages



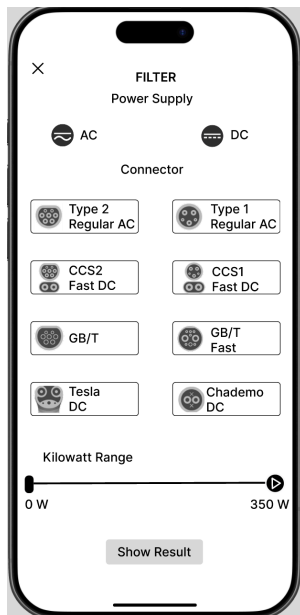
Home Page



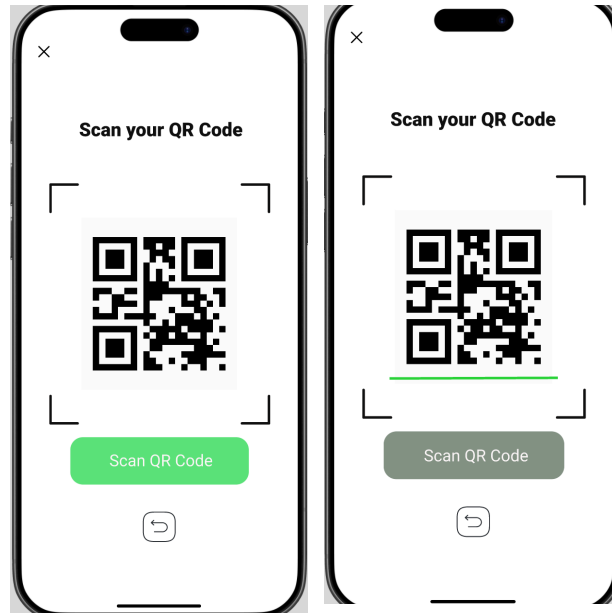
Search Page



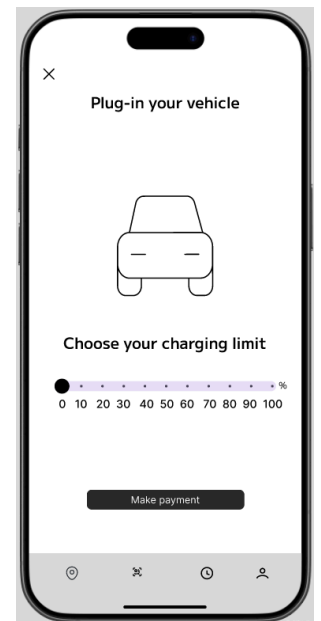
Direct to map page



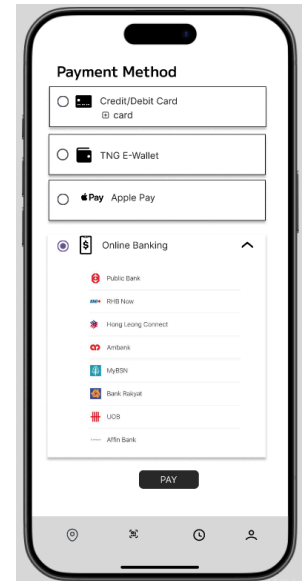
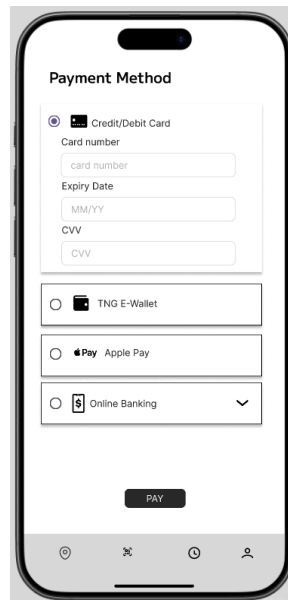
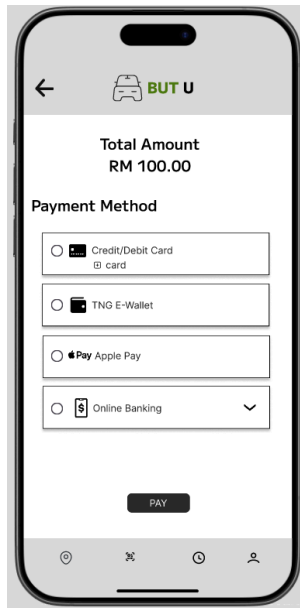
Filter Page



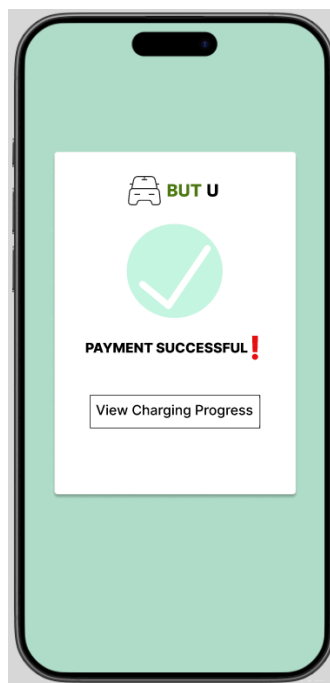
QR Scanning Pages



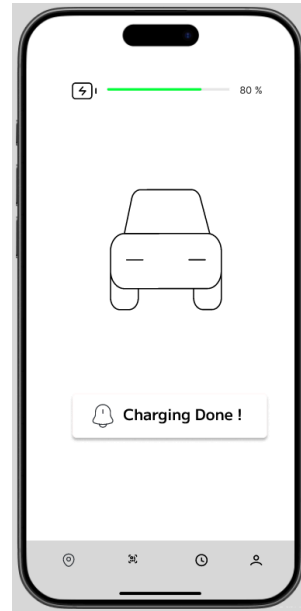
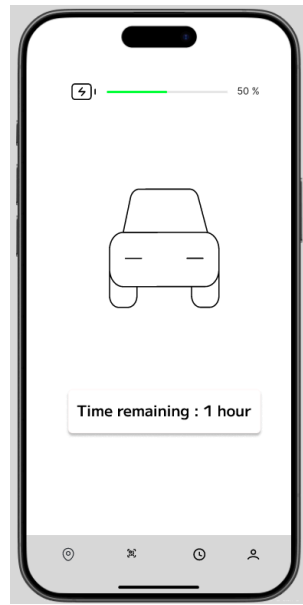
Limit and start charging page



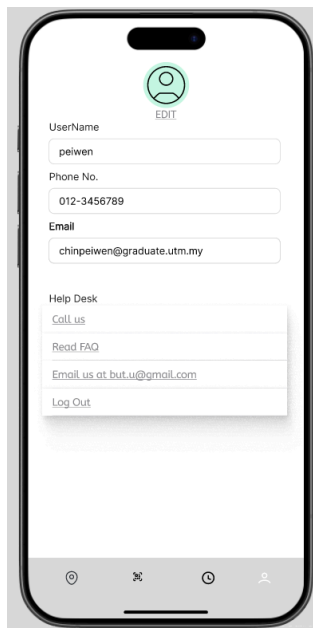
Payment Pages



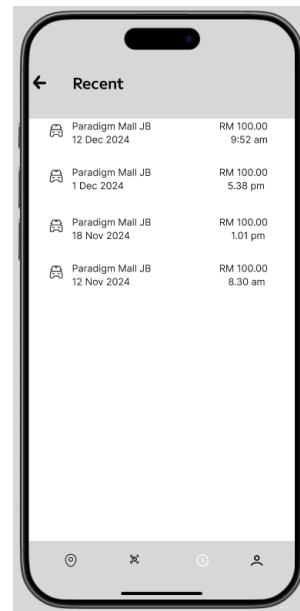
Payment Success Page



Charging Progress Pages



Profile Page



History Page

Briefing notes – prepared by {Koo Xuan}

We are a group of students from UTM which is currently developing a mobile application called **BUT U**. This app is designed to assist electric vehicle (EV) users in locating and utilizing charging stations efficiently. This project aligns with Sustainable Development Goals (SDGs), particularly **SDG 7 (Affordable and Clean Energy)** and **SDG 13 (Climate Action)**. Our app aims to provide a seamless and user-friendly experience for EV owners, fleet managers, and travelers to access EV charging infrastructure.

So we hope that “user” could give us about 30 minutes. And this interview is to gather insights, feedback, and preferences from “users” of the BUT U app. Your input will help us improve the app's usability, functionality, and relevance to meet user expectations effectively.

Now, I'm going to introduce our app which includes 3 main features which is **Finding Charging Stations** that locate nearby EV charging stations based on your location or desired route. Next, **Make Payment** for the station, this app allows users to pay for charging sessions using various payment methods, including e-wallets, credit cards, and loyalty points. Last, **Monitoring Charging Process** which allows users to view real-time updates on their EV's charging progress remotely.

Testing with users

Introduction

{Embedded/link video on User 1}: <https://youtu.be/KIACw24o3gk>

{Embedded/link video on User 2}: <https://youtu.be/IbgEvZNRD88>

{Embedded/link video on User 3}: <https://youtu.be/60A-oCOB2HM>

Task 1

{Embedded/link video on User 1}: <https://youtu.be/oEkT-4qRkkU>

{Embedded/link video on User 2}: <https://youtu.be/aa2XT4CDCII>

{Embedded/link video on User 3}: <https://youtu.be/mB61p9Emk1Y>

Task 2

{Embedded/link video on User 1}: <https://youtu.be/svULwEz2uzE>

{Embedded/link video on User 2}: <https://youtu.be/yJRxSyzerFk>

{Embedded/link video on User 3}: <https://youtu.be/bG0StCLZN-A>

Task 3

{Embedded/link video on User 1}: <https://youtu.be/lm85W-MqPP8>

{Embedded/link video on User 2}: https://youtu.be/DT_TVzXwflw

{Embedded/link video on User 3}: <https://youtu.be/rf2kTb4u3F4>

Interview

{Embedded/link video on User 1}: <https://youtu.be/rPNuGZiwJ-A>

{Embedded/link video on User 2}: <https://youtu.be/aZ-rgp8H87c>

{Embedded/link video on User 3}: <https://youtu.be/gmYrIyu00OU>

Observations

User 1 (EV Owner/Driver) - Prepared by Ling Yu Qian

Observation

The electric vehicle owner primarily charges her EV at home using a dedicated charging port, avoiding public charging unless traveling long distances. She noted some limitations in the app's performance, particularly related to charging limit functionalities, and believes improvements could enhance the user experience. The user appreciated the app's ability to simplify locating charging stations and recommended it to others for its ease of use and convenience.

Results of the Interview

The electric vehicle owner rated the app 7/10, citing some limitations in its functionality, particularly with the charging limit feature, which she suggested could be improved. She primarily charges her EV at home using a dedicated port and rarely uses public charging stations unless traveling long distances. Despite the limitations, she found the app highly useful for simplifying the process of locating charging stations and determining availability, which significantly aids in travel planning. She recommended the app to other EV owners, emphasizing its convenience and value for both experienced and new users navigating charging infrastructure.

User 2 (Manager) - Prepared by Ling Yu Qian

Observation

The electric vehicle owner found the system user-friendly and intuitive, stating that they could navigate and use it without guidance. They particularly appreciated the ability to monitor EV charging progress remotely, emphasizing the convenience of receiving live updates on charging status as well as notifications when charging is complete. These features provided flexibility to check progress actively or passively, enhancing the overall user experience. The user also noted that the system is both useful and convenient for managing EV charging.

Result of Interview

The electric vehicle owner rated the system 7/10, highlighting its user-friendly design and intuitive functionality, which allowed them to navigate and use it without prior guidance. The ability to monitor EV charging progress remotely was deemed very convenient, as the combination of live updates and notifications provided flexibility to actively check progress or receive alerts when charging was complete. These features enhanced the overall user experience and made the system versatile for different scenarios. The user recommended the system to others for its convenience and utility in simplifying EV charging management, while also

suggesting that improvements could be made to fully meet user expectations and deliver an even better experience.

User 3 (Traveler) - Prepared by Ling Yu Qian

Observation

The traveler shared her experience using the system. She mentioned challenges during long-distance travel with her electric vehicle, such as difficulties accessing charging infrastructure and receiving assistance on highways. Despite these limitations, the traveler appreciated the system's ability to locate nearby charging stations based on her current location, finding it highly useful for planning her trips. However, she expressed uncertainty about whether she could fully trust and utilize the system's data for her travel needs.

Results of the Interview

The traveler rated the app 8/10, expressing appreciation for its ability to locate nearby charging stations based on real-time location, a feature she found crucial for planning long-distance trips. She noted the app's potential to alleviate common frustrations faced during travel, such as the lack of accessible charging infrastructure and reliable assistance on highways. However, she highlighted a need for improvements in the system's reliability to ensure users can fully utilize the data provided. Overall, the traveler acknowledged the app's usefulness in addressing travel challenges and emphasized its importance for stress-free long-distance journeys. Highlighted the importance of the system for addressing long-distance travel challenges with electric vehicles in high way , particularly in areas with limited infrastructure. So she no need call company to charge her vehicle.

Findings – prepared by {Koo Xuan}

User 1: EV Car Owner

Usability Problem

During the testing phase, the EV car owner found it inconvenient to check the station's real-time queue, leading to unexpected waiting times at busy charging stations.

Solution

Integrate a queue management system within the app that shows the current number of users waiting at each station. Provide an option for users to join a virtual queue and receive notifications when their turn is approaching.

User 2: Company Representative

Usability Problem

The company representative found it challenging to manage multiple company EVs due to the lack of a feature for monitoring multiple charging sessions simultaneously.

Solution

Add a "Fleet Management" feature that allows users to track multiple vehicles and their charging statuses in one consolidated view. Include options to assign specific vehicles to charging stations and set notifications for charging completion for each vehicle.

User 3: Traveler

Usability Problem

The traveler faced difficulty finding charging stations along their planned route, as the app lacked an integrated route planning feature.

Solution

Introduce a route planning feature that allows users to input their destination and view charging stations along the way. Highlight stations that are most convenient based on travel distance, charger availability, and estimated arrival times.